

CHAPTER 9 : ENVIRONMENTAL MANAGEMENT

Topic:-

Establish a vermi compost project in your school under the guidance of your teachers.

Introduction:-

Modern human activities generate enormous quantities of solid organic waste every single day. In schools, organic waste accumulates continually from garden dry leaf litter, lawn grass cuttings, pruned twigs, and leftover food scraps from the student midday meal canteen. If dumped in open garbage heaps, this organic waste rots anaerobically, emitting foul odors and attracting disease-carrying flies and rodents.

Nature, however, offers an extraordinary biological recycling solution called **Vermicomposting**. Vermicomposting is the scientific process of converting biodegradable organic waste into nutrient-rich, dark, friable organic manure using specialized species of earthworms. Often revered as the "Farmer's Best Friend", earthworms consume organic debris, digest it with beneficial gut microflora, and excrete granular casts known as **vermicast**.

Under the guidance of our science teachers and school gardener, our class established an operational vermicompost unit in our school botanical garden. We utilized discarded school crates, organic canteen wastes, and the red earthworm species *Eisenia fetida*. We monitored the pit over eight weeks, harvesting superior organic fertilizer for our school flowerbeds.

❖ Core Advantages of Vermicomposting:-

- Produces "Black Gold" organic manure rich in Nitrogen, Phosphorus, Potassium, and growth hormones.
- Eliminates organic waste accumulation in school premises with zero chemical pollutants.
- Restores natural soil micro-ecosystems and reduces dependence on synthetic fertilizers.

Aim & Vermicompost Unit Setup:-

Aim: To set up, maintain, and monitor a functional vermicompost bed in the school garden using organic kitchen/garden waste and red earthworms under teacher guidance.

❖ *Objectives of the Project:-*

- To construct a well-aerated vermicomposting bed or wooden crate unit in the shaded school garden.
- To procure and introduce healthy red worm cultures (*Eisenia fetida*) into moist organic bedding.
- To feed worms chopped vegetable canteen peels, tea waste, and semi-decomposed dry leaves weekly.
- To maintain optimum moisture, aeration, and temperature, harvesting matured vermicompost after 60 days.

❖ *Materials & Supplies Assembled:-*

- Wooden rectangular crate / masonry pit (size: 1.2 m × 0.9 m × 0.6 m) with bottom drainage holes.
- Coarse gravel, river sand, shredded cardboard, dry leaves, and coconut coir for basal bedding.
- 1 kg of active red wiggler earthworms (*Eisenia fetida*) sourced from local agriculture Krishi Vigyan Kendra.
- Water sprinkler can, hand trowel, jute gunny bags for covering, and wire mesh sieve.



Step-by-Step Setup Procedure:-

The construction and commissioning of our school vermicompost unit followed this systematic procedure:

❖ *Construction & Inoculation Steps:-*

- **Step 1 (Site Selection):** Selected a cool, shaded corner in the school backyard shielded from harsh afternoon sun and torrential rains by a thick neem tree canopy.
- **Step 2 (Basal Drainage Layer):** Spread a 5 cm base layer of broken brick pieces and coarse river sand at the bottom of the wooden crate to ensure excess water drains without suffocating the worms.
- **Step 3 (Bedding Preparation):** Placed a 10 cm layer of moist coconut coir, shredded egg cartons, and partially aged cow dung. Sprinkled water until damp like a wrung-out sponge (50%–60% moisture).
- **Step 4 (Earthworm Inoculation):** Gently released 1 kg of red wiggler earthworms (*Eisenia fetida*) onto the bedding surface. The worms naturally burrowed downward into the dark, moist substrate within 15 minutes.
- **Step 5 (Feeding & Covering):** Added finely chopped canteen vegetable scraps and fallen leaves in a 5 cm layer. Covered the crate with a moist jute gunny cloth to ensure complete darkness and conserve moisture.

Observations & Weekly Progress Table:-

Over an eight-week monitoring period, our team inspected the pit weekly, recording moisture, temperature, waste breakdown, and worm activity:

Timeline	Waste Condition	Pit Temp (°C)	Moisture Status	Earthworm Activity Observed
Week 1	Raw fresh vegetable peels	25°C	Optimal (60%)	Acclimatizing in basal bedding
Week 2	Softening, mild mold growth	27°C	Moderate	Actively feeding at upper boundary
Week 4	50% volume reduced, darkening	26°C	Sprinkled water	Worms multiplying; juvenile cocoons seen
Week 6	Scraps nearly unrecognizable	24°C	Good dampness	Abundant worm castings accumulating
Week 8	Granular, dark tea-powder texture	23°C	Reduced watering	Vermicompost ready for sieving

The final product was completely odorless, dark brown, and granular with an earthy forest smell.

Scientific Analysis & Digestive Mechanism:-

Analyzing the biological transformation reveals why earthworm composting is vastly superior to passive rotting:

❖ *Biological Mechanisms in Vermicomposting:-*

- **Gizzard Grinding & Microflora Action:** Earthworms lack teeth. Ingested food particles pass into their muscular gizzard, where tiny mineral grains grind the waste. Their intestine secretes digestive enzymes (cellulase, protease, amylase), while millions of symbiotic gut microbes break down cellulose and lignin.
- **Nutrient Enrichment in Vermicast:** Scientific agricultural tests reveal that worm castings contain 5 times more available Nitrogen, 7 times more Phosphorus, 11 times more Potassium, and 3 times more Magnesium than ordinary garden topsoil!
- **Presence of Plant Growth Regulators:** Vermicompost contains natural plant auxins, gibberellins, and humic acids that stimulate rapid root development and enhance plant resistance to fungal pathogens.

The red worm *Eisenia fetida* was selected because it is an epigeic surface-dwelling detritivore capable of consuming its own body weight in organic matter daily.

Harvesting & Garden Application:-

After 60 days, we harvested and applied our school-made organic vermicompost:

❖ Harvesting Protocol:-

- **Withholding Water:** Stopped sprinkling water 3 days prior to harvesting. The top layer dried slightly, prompting worms to migrate down to the moist lower bedding.
- **Pyramid Heap Method:** Scooped the compost into cone-shaped heaps in bright light. Light-sensitive worms burrowed to the center, allowing us to collect clean castings from the surface.
- **Sieving & Storage:** Passed the compost through a 3 mm wire mesh sieve. Returned captured earthworms and cocoons back to the pit for the next batch cycle.

❖ School Garden Applications & Results:-

- Harvested 35 kg of premium vermicompost from approximately 90 kg of school organic scraps.
- Applied vermicompost to school rose bushes, marigold beds, and potted hibiscus shrubs.
- Within four weeks, flowering shrubs exhibited deeper green foliage and 40% more vibrant blooms.

What We Learned, Conclusion & References:-

❖ What We Learned From This Project:-

- Mastered practical techniques of organic waste recycling and earthworm culture management.
- Learned the vital role of detritivores in maintaining ecological nutrient loops and soil fertility.
- Developed teamwork, manual craftsmanship, and environmental stewardship skills.

❖ Project Conclusion:-

Our school vermicomposting project converted troublesome canteen and garden waste into rich, valuable organic manure. The initiative achieved zero-waste management in our school garden while providing free organic fertilizer for campus greenery. Vermicomposting proves that simple biological innovations can turn environmental waste into agricultural treasure.

❖ Bibliography & References:-

- Maharashtra State Bureau of Textbook Production (Balbharati) – Standard 9 Science & Technology (Chapter 9: Environmental Management).
- Krishi Vigyan Kendra (KVK), Baramati – *Technical Manual on Vermiculture and Vermicomposting*.
- Sultan Ismail – *The Earthworm Book*, Other India Press, Mapusa, Goa.